

ALFAA47362

## Platinum, nominally 18%, cobalt, nominally 1%, nickel, nominally 1% on durable carbon support

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 载铂催化剂(18%铂, 1%钴, 1%镍负载在碳上)  
 Platinum, nominally 18%, cobalt, nominally 1%, nickel, nominally 1% on durable carbon support

**Cat No. :** 47362

**Supplier** Avocado Research Chemicals Ltd.  
 (Part of Thermo Fisher Scientific)  
 Shore Road, Heysham  
 Lancashire, LA3 2XY,  
 United Kingdom  
 Office Tel: +44 (0) 1524 850506  
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**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
 Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
 Powder Solid

**Appearance**  
 Black

**Odor**  
 No information available

#### Emergency Overview

May cause an allergic skin reaction. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Suspected of causing genetic defects. May cause cancer. May damage fertility or the unborn child. May cause damage to organs through prolonged or repeated exposure.

#### Classification of the substance or mixture

|                                                      |             |
|------------------------------------------------------|-------------|
| Respiratory Sensitization                            | Category 1  |
| Skin Sensitization                                   | Category 1  |
| Germ Cell Mutagenicity                               | Category 2  |
| Carcinogenicity                                      | Category 1B |
| Reproductive Toxicity                                | Category 1B |
| Specific target organ toxicity - (repeated exposure) | Category 2  |

#### Label Elements

Platinum, nominally 18%, cobalt, nominally 1%, nickel, nominally 1% on durable carbon support

**Signal Word****Danger****Hazard Statements**

H317 - May cause an allergic skin reaction  
H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled  
H341 - Suspected of causing genetic defects  
H350 - May cause cancer  
H373 - May cause damage to organs through prolonged or repeated exposure  
H360 - May damage fertility or the unborn child

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves  
P284 - In case of inadequate ventilation wear respiratory protection

**Response**

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P341 - IF INHALED: If breathing is difficult, remove victim to fresh air and keep at rest in a position comfortable for breathing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P342 + P311 - If experiencing respiratory symptoms: Call a POISON CENTER or doctor  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

May cause an allergic skin reaction. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Suspected of causing genetic defects. May cause cancer. May damage fertility or the unborn child. May cause damage to organs through prolonged or repeated exposure.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component      | CAS No    | Weight % |
|----------------|-----------|----------|
| Carbon black   | 1333-86-4 | 80       |
| Platinum       | 7440-06-4 | 18       |
| Nickel         | 7440-02-0 | 1        |
| Cobalt, powder | 7440-48-4 | 1        |

**SECTION 4. FIRST AID MEASURES**

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**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable. May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

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## SECTION 7. HANDLING AND STORAGE

**Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage**

Keep container tightly closed in a dry and well-ventilated place.

**Specific Use(s)**

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

**Control Parameters**

| Component      | China                                                      | Taiwan                      | Thailand                   | Hong Kong                   |
|----------------|------------------------------------------------------------|-----------------------------|----------------------------|-----------------------------|
| Carbon black   | TWA: 4 mg/m <sup>3</sup>                                   | TWA: 3.5 mg/m <sup>3</sup>  |                            | TWA: 3.5 mg/m <sup>3</sup>  |
| Platinum       | -                                                          | TWA: 1 mg/m <sup>3</sup>    |                            | TWA: 1 mg/m <sup>3</sup>    |
| Nickel         | TWA: 1 mg/m <sup>3</sup>                                   | TWA: 1 mg/m <sup>3</sup>    | TWA: 1 mg/m <sup>3</sup>   | TWA: 1.5 mg/m <sup>3</sup>  |
| Cobalt, powder | TWA: 0.05 mg/m <sup>3</sup><br>STEL: 0.1 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> |

| Component      | ACGIH TLV                   | OSHA PEL                                                            | NIOSH                                                                                    | The United Kingdom                                                                   | European Union |
|----------------|-----------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------|
| Carbon black   | TWA: 3 mg/m <sup>3</sup>    | (Vacated) TWA: 3.5 mg/m <sup>3</sup><br>TWA: 3.5 mg/m <sup>3</sup>  | IDLH: 1750 mg/m <sup>3</sup><br>TWA: 3.5 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | STEL: 7 mg/m <sup>3</sup> 15 min<br>TWA: 3.5 mg/m <sup>3</sup> 8 hr                  |                |
| Platinum       | TWA: 1 mg/m <sup>3</sup>    | (Vacated) TWA: 1 mg/m <sup>3</sup>                                  | TWA: 1 mg/m <sup>3</sup>                                                                 | STEL: 15 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr                   |                |
| Nickel         | TWA: 1.5 mg/m <sup>3</sup>  | (Vacated) TWA: 1 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup>      | IDLH: 10 mg/m <sup>3</sup><br>TWA: 0.015 mg/m <sup>3</sup>                               | STEL: 1.5 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr<br>Skin        |                |
| Cobalt, powder | TWA: 0.02 mg/m <sup>3</sup> | (Vacated) TWA: 0.05 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | IDLH: 20 mg/m <sup>3</sup><br>TWA: 0.05 mg/m <sup>3</sup>                                | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Resp. Sens. |                |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust MDHS30/2 Cobalt and cobalt compounds in air Laboratory method using flame atomic absorption spectrometry MDHS46/2 Platinum metal and soluble platinum compounds in air Laboratory method using electrothermal atomic absorption spectrometry or inductively coupled plasma-mass spectrometry MDHS42/2 Nickel and inorganic compounds of nickel in air (except nickel carbonyl) Laboratory method using flame atomic absorption spectrometry or electrothermal atomic absorption spectrometry

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

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**Hand Protection**

Protective gloves

| Glove material              | Breakthrough time                    | Glove thickness | EU standard | Glove comments        |
|-----------------------------|--------------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber<br>Viton (R) | See manufacturers<br>recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use**

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371 Type A Brown**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES****Appearance**

Black

**Physical State**

Powder Solid

**Odor**

No information available

**Odor Threshold**

No data available

**pH**

No information available

**Melting Point/Range**

No data available

**Softening Point**

No data available

**Boiling Point/Range**

No information available

**Flash Point**

No information available

**Method -** No information available**Evaporation Rate**

Not applicable

Solid

**Flammability (solid,gas)**

No information available

**Explosion Limits**

No data available

**Vapor Pressure**

No data available

**Vapor Density**

Not applicable

Solid

**Specific Gravity / Density**

No data available

**Bulk Density**

No data available

**Water Solubility**

No information available

**Solubility in other solvents**

No information available

**Partition Coefficient (n-octanol/water)**

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|                           |                          |       |
|---------------------------|--------------------------|-------|
| Autoignition Temperature  | Not applicable           |       |
| Decomposition Temperature | No data available        |       |
| Viscosity                 | Not applicable           | Solid |
| Explosive Properties      | No information available |       |
| Oxidizing Properties      | No information available |       |

## SECTION 10. STABILITY AND REACTIVITY

**Stability** Stable under normal conditions.

**Hazardous Reactions** None under normal processing.  
**Hazardous Polymerization** No information available.

**Conditions to Avoid** None known.

**Materials to avoid** No information available.

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Platinum oxide. Cobalt oxides. Nickel oxides.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

(a) acute toxicity;  
**Toxicology data for the components**

| Component      | LD50 Oral                  | LD50 Dermal              | LC50 Inhalation                          |
|----------------|----------------------------|--------------------------|------------------------------------------|
| Carbon black   | LD50 > 15400 mg/kg ( Rat ) | LD50 > 3 g/kg ( Rabbit ) | LC50 > 4.6 mg/m <sup>3</sup> ( Rat ) 4 h |
| Nickel         | LD50 > 9000 mg/kg ( Rat )  |                          | LC50 > 10.2 mg/L ( Rat ) 1 h             |
| Cobalt, powder | LD50 = 6171 mg/kg ( Rat )  |                          | LC50 < 0.05 mg/L ( Rat ) 4 h             |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
**Respiratory** Category 1  
**Skin** Category 1  
 No information available

(e) germ cell mutagenicity; Category 2

(f) carcinogenicity; Category 1B  
 The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component      | EU           | UK | Germany | IARC     |
|----------------|--------------|----|---------|----------|
| Carbon black   |              |    | Cat. 3B | Group 2B |
| Nickel         |              |    | Cat. 1  | Group 2B |
| Cobalt, powder | Carc Cat. 1B |    | Cat. 2  | Group 2A |

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(g) reproductive toxicity; Category 1B

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 2

Route of exposure Inhalation  
Target Organs Lungs.(j) aspiration hazard; Not applicable  
SolidSymptoms / effects, both acute and delayed Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing:  
Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## SECTION 12. ECOLOGICAL INFORMATION

## Ecotoxicity effects

Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component      | Freshwater Fish                                                                                                                                     | Water Flea                                                                          | Freshwater Algae                                                                                                                   | Microtox |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------|
| Carbon black   |                                                                                                                                                     | EC50: > 5600 mg/L, 24h (Daphnia magna)                                              |                                                                                                                                    |          |
| Nickel         | LC50: > 100 mg/L, 96h (Brachydanio rerio)<br>LC50: = 1.3 mg/L, 96h semi-static (Cyprinus carpio)<br>LC50: = 10.4 mg/L, 96h static (Cyprinus carpio) | EC50: = 1 mg/L, 48h Static (Daphnia magna)<br>EC50: > 100 mg/L, 48h (Daphnia magna) | EC50: 0.174 - 0.311 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: = 0.18 mg/L, 72h (Pseudokirchneriella subcapitata) |          |
| Cobalt, powder | LC50: > 100 mg/L, 96h static (Brachydanio rerio)                                                                                                    |                                                                                     |                                                                                                                                    |          |

## Persistence and Degradability

Persistence  
Degradation in sewage treatment plantProduct contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
May persist.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative Potential

Product has a high potential to bioconcentrate

## Mobility in soil

No information available

Endocrine Disruptor Information  
Persistent Organic Pollutant  
Ozone Depletion PotentialThis product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

Platinum, nominally 18%, cobalt, nominally 1%, nickel, nominally 1% on durable carbon support

|                                            |                                                                                                                                                                        |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                              |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.                   |

**SECTION 14. TRANSPORT INFORMATION**

**Road and Rail Transport** Not Regulated

**IMDG/IMO** Not regulated

**IATA** Not regulated

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed.

| Component      | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|----------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Carbon black   | -                                                   | -                                       | X    | X     | 215-609-9 | X    | X   | X     | X    | X    | X    | X        |
| Platinum       | -                                                   | -                                       | X    | X     | 231-116-1 | X    | X   | X     | X    |      | X    | KE-28808 |
| Nickel         | -                                                   | -                                       | X    | X     | 231-111-4 | X    | X   | X     | X    |      | X    | KE-25818 |
| Cobalt, powder | -                                                   | -                                       | X    | X     | 231-158-0 | X    | X   | X     | X    |      | X    | KE-06060 |

**National Regulations****SECTION 16. OTHER INFORMATION**

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 16-Sep-2019  
**Revision Date** 12-May-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.



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**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

**Environmental hazards**

Calculation method

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**